

3. WATER JUG PROBLEM

Aim:

To develop a Python program to solve the Water Jug Problem and obtain the required quantity of water using two water jugs.

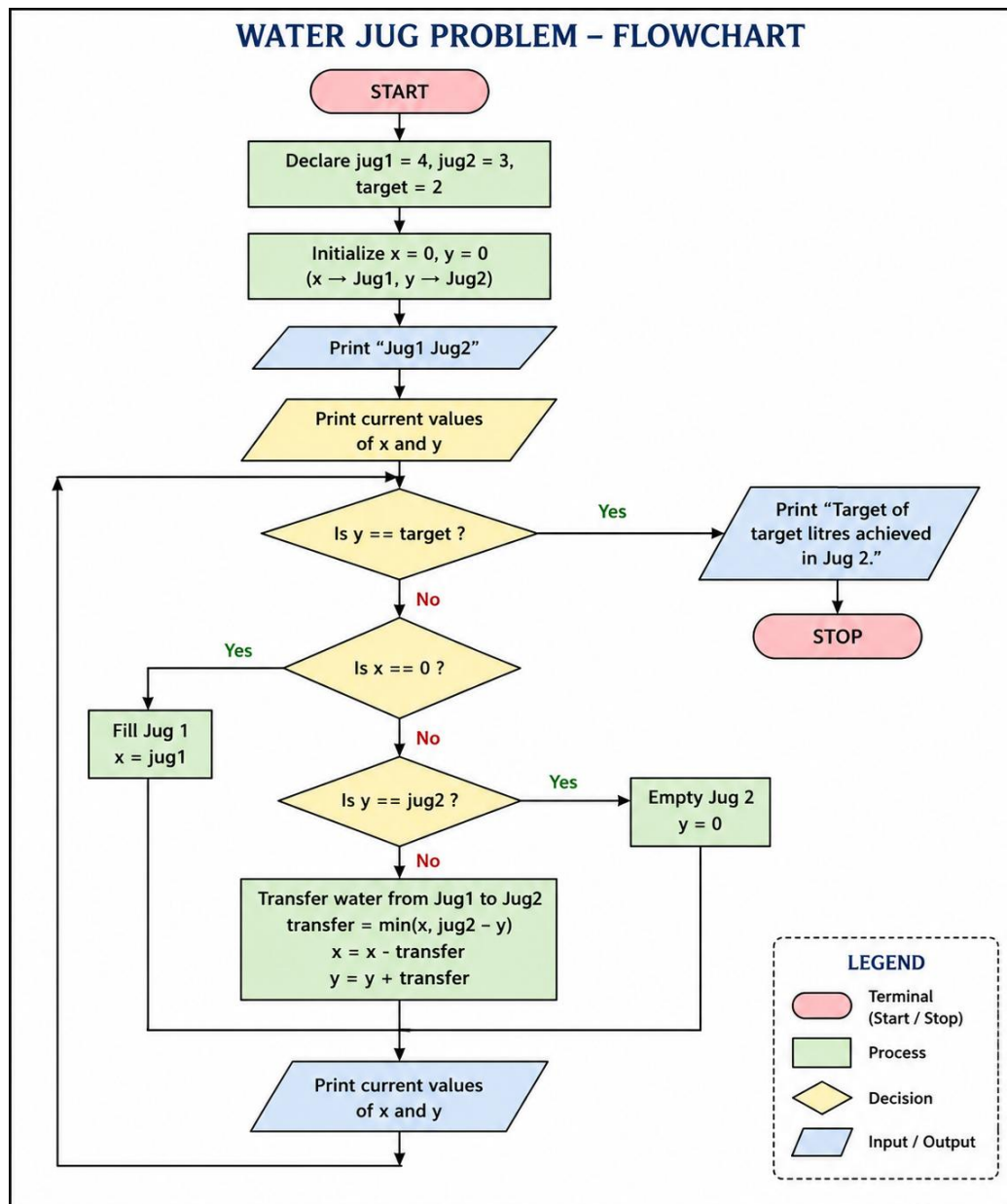
Objective :

- To understand the Water Jug Problem.
- To simulate the filling, emptying, and transferring of water between two jugs.
- To obtain the required target quantity of water.
- To display each intermediate state until the target is reached.

Algorithm:

1. Start.
2. Define the capacities of Jug 1 and Jug 2.
3. Specify the target quantity of water.
4. Initially, both jugs are empty.
5. Repeat until the target quantity is obtained:
 - If Jug 1 is empty, fill it completely.
 - Else if Jug 2 is full, empty Jug 2.
 - Otherwise, pour water from Jug 1 into Jug 2 until Jug 2 is full or Jug 1 becomes empty.
6. Display the amount of water in both jugs after every operation.
7. When the target quantity is obtained, display the success message.
8. Stop.

Flowchart :



Code:

jug1 = 4

jug2 = 3

target = 2

x = 0

```

y = 0
print("Jug1 Jug2")
while y != target:
    if x == 0:
        x = jug1
    elif y == jug2:
        y = 0
    else:
        transfer = min(x, jug2 - y)
        x -= transfer
        y += transfer
    print(x, " ", y)
print("\nTarget of", target, "litres achieved in Jug 2.")

```

Output:

```

Output
Jug1 Jug2
4     0
1     3
1     0
0     1
4     1
2     3
2     0
0     2

Target of 2 litres achieved in Jug 2.

=== Code Execution Successful ===

```

Result :

The Water Jug Problem was successfully implemented using Python. The program simulated the operations of filling, emptying, and transferring water between two jugs until the required target quantity of 2 litres was obtained successfully.